

ICCAD '20

GAMMA

Sheng-Chun Kao and Tushar Krishna
Georgia Institute of Technology
November 2020

Outline

1. Motivation

2. Design

3. Evaluation

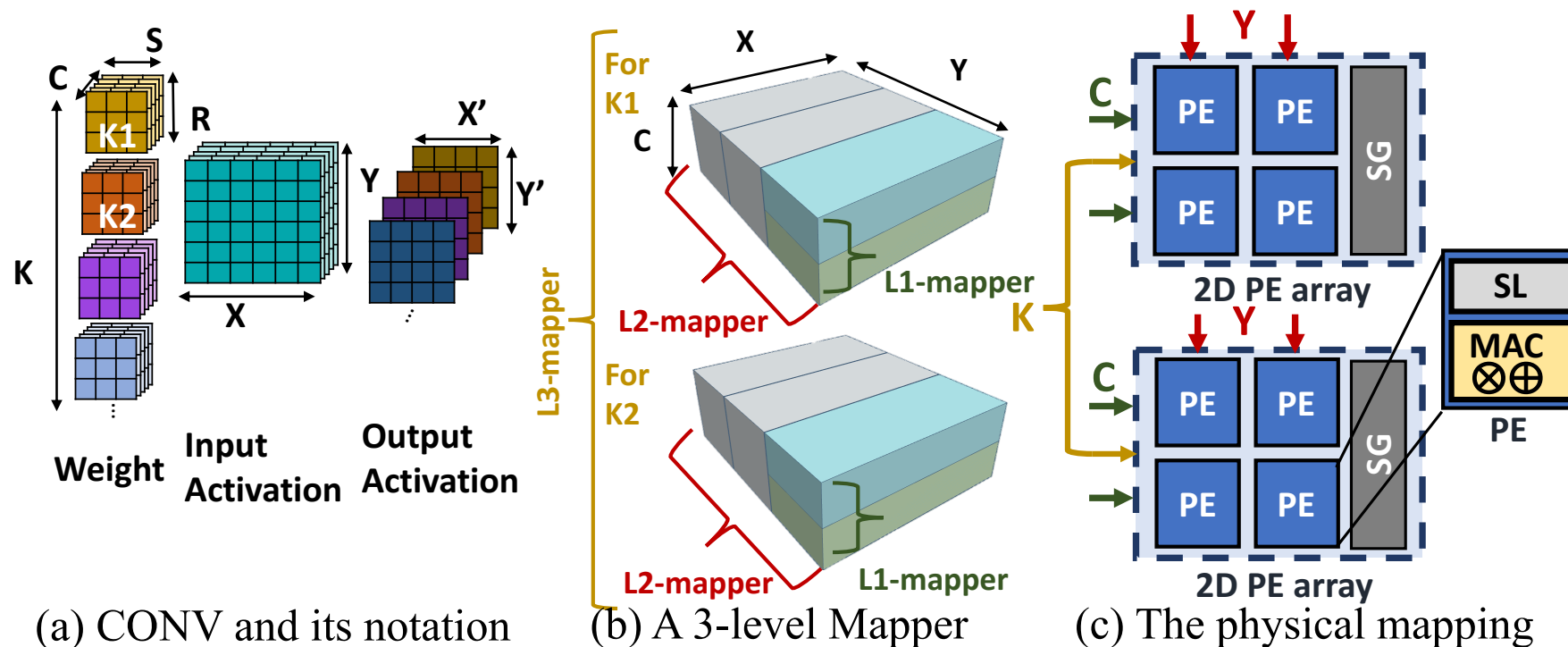
4. Takeaways

What Mapping Controls

3

Mapping: one schedule jointly chooses parallelism, computation order, and tile sizes.

Consequence: those choices set reuse and memory traffic, so no fixed dataflow stays efficient across early, middle, and late CNN layers.



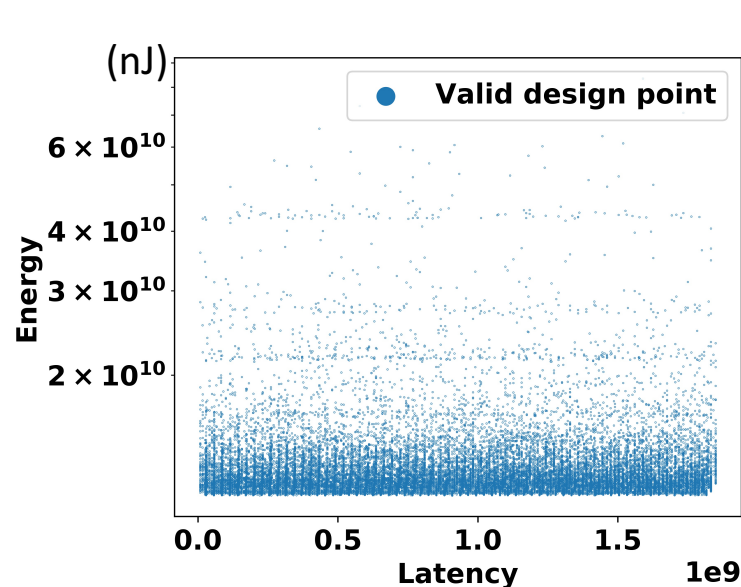
CONV dimensions, hierarchical mapper, and physical mapping

The Search Space Is the Barrier

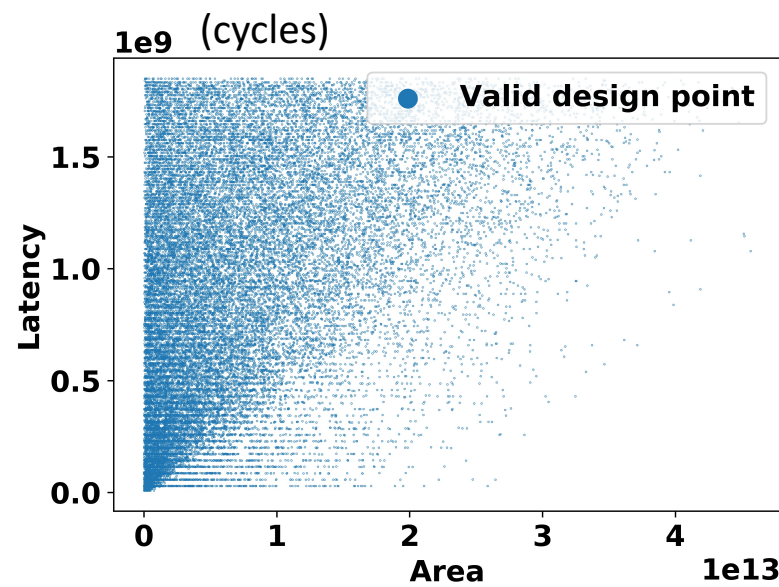
4

Motivation: even one VGG16 layer spans orders of magnitude in latency and energy under different valid mappings.

Scale: layer 2 is already $O(10^{12})$ at one level and $O(10^{36})$ at three levels, so brute force and random search are poor fits.



(a) Energy-Latency



(b) Latency-Area

	Min	Max
Latency (cycles)	8.18E+06	1.85E+09
Energy (nJ)	1.12E+10	8.34E+10
Power (mW)	1.52E+05	2.24E+09
Area (μm^2)	1.77E+09	4.57E+13
SL size (Bytes)	2.62E+02	9.90E+06
SG size (Bytes)	2.82E+03	1.54E+07

(c)HW perf. statistics

10K random mappings for one VGG16 layer cover radically different hardware outcomes

Outline

1. Motivation
- 2. Design**
3. Evaluation
4. Takeaways

Core claim: search the complete map space instead of fixing the number of parallel levels in advance.

Representation: the genome carries parallel dimension, compute order, and tile sizes in one object.

Operators: reorder, growth, and aging let the search move across variable-depth mappings without a separate learned encoder.

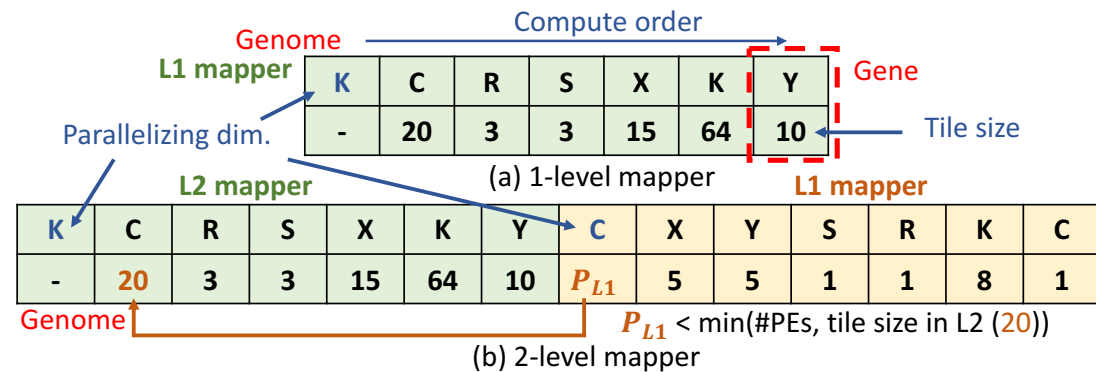
Evaluation loop: MAESTRO scores each generation, while invalid mappings receive negative-infinity fitness.

Encoding the Full Map Space

7

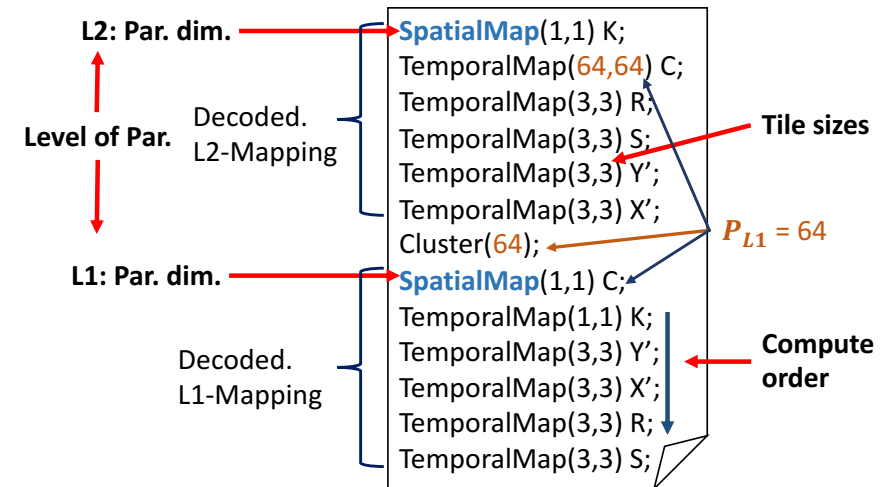
Genome: a 1-level mapper is encoded as 7 gene pairs; multi-level mappings concatenate mapper levels.

Decode: each level becomes one Cluster in MAESTRO, with the first gene pair as SpatialMap and the rest as TemporalMap.



L2-Mapper							L1-Mapper						
K	K	C	R	S	X	Y	C	C	K	Y	X	R	S
-	1	64	3	3	3	3	64	1	1	3	3	3	3

(a) Genome of 2-level mapper



Decoded mapping in MAESTRO form

Flexible genome for 1-level and 2-level mappers

Why Fixed-Length Baselines Break

8

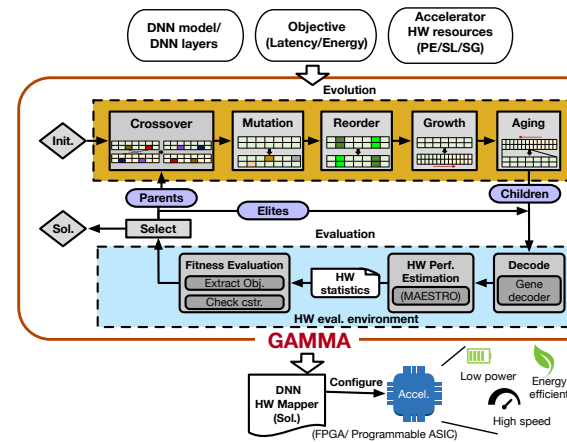
Baseline limit: most generic optimizers assume fixed-length inputs, which freezes the number of parallel levels.

Consequence: once mapping depth is fixed in advance, richer accelerators need separately hand-built search spaces instead of one flexible search loop.

Variable-Depth Operators

GAMMA move: reorder changes loop order, while growth and aging add or remove a mapper level.

Result: one search loop can span s_1 to s_3 instead of hand-building a different optimizer input for each depth.



(a) Structure and algorithm flow

Optimization methods	Description	Mapping Space Aspects			
		Tile Strat.	Com. Ord.	Par. Dim.	Par. Level
Baseline Methods	Defined a <i>Par. Level</i> . Encode <i>Tile Strat.</i> , <i>Com. Ord.</i> , <i>Par. Dim.</i> into fixed length inputs, and evolve them with algorithms.	✓	✓	✓	✗
GAMMA	Crossover	✓			
	Mutation	✓		✓	
	Reorder		✓		
	Growth				✓
	Aging	✓	✓		✓
	Summary	✓	✓	✓	✓

(b) The summary of evolution in GAMMA

Workflow plus operator summary for GAMMA

Outline

- 1. Motivation
- 2. Design
- 3. Evaluation**
- 4. Takeaways

Experimental Setup

11

Coverage: 5 CNNs, 2 hardware budgets, 3 accelerator organizations, and latency or energy objectives.

Models: VGG16, MobileNet-V2, ResNet-18, ResNet-50, and MnasNet.

Question: does one search loop stay effective as the accelerator moves from fixed s_1 to flexible s_3 ?

	# PEs	SL size (on-chip)	SG size (on-chip)
Edge Platform	168	512B	108KB
Cloud Platform	65,536	4MiB	24MiB

Edge and Cloud hardware budgets

System	Description
S1:Fixed 2D Accel	Accelerator with 2D PE array (row x col) with fixed aspect ratio. This fixes P_{L1} to number of rows, and level of parallelism to 2. Example: TPU, NVDLA.
S2: Flexible 2D Accel	Accelerator with 2D PE array with flexible aspect ratio. The PE array can be configured to flexible 2D shape, which relax the choices of P_{L1} . We also relax the level of parallelism to be 1 or 2. Example: Eyeriss, Eyeriss_v2.
S3: Scale-out 2D Accel	Accelerator scaling out 2D structure. Accelerator comprises multiple 2D PE array instances, enabling parallelism across PE arrays. The level of parallelism is either 2 or 3. L1 and L2 mappers map within and L3 mapper maps across the PE arrays. Example: Simba, Tetris.

Target systems S1, S2, and S3

Baselines and Search Budget

12

Generic optimizers: RS, GA, DE, (1+lambda)-ES, CMA-ES, TBPSA, PSO, and pPortfolio.

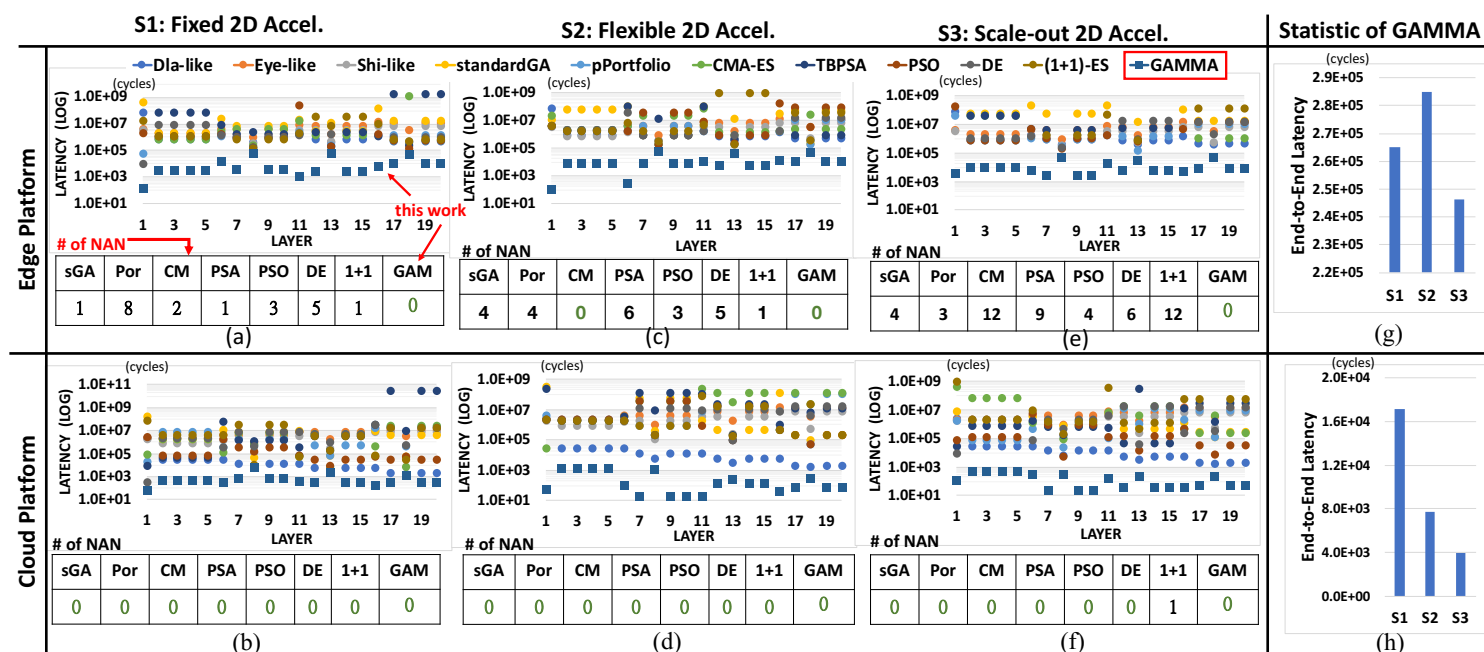
Fixed dataflows: NVDLA-, Eyeriss-, and ShiDianNao-like mappings test whether GAMMA only beats weak baselines.

Budget: every method gets 10K samples; GAMMA uses population 200, 50 generations, and 0.5 evolution rates.

Validity pressure: mappings that exceed SL or SG are invalid, so some baselines fail before they become competitive.

Main result: GAMMA both finds valid mappings and beats generic optimizers when constraints are tight or the map space becomes more flexible.

Anchor numbers: S1 Edge 224x-440x, S2 Edge 209x-1,035x, S2 Edge energy 11x-36x, and S3 Edge 241x-644x better than alternatives.



Optimization results across S1, S2, and S3 on Edge and Cloud

The Learned Mappings Match Layer Shape

14

Interpretability: GAMMA does not discover arbitrary schedules; it shifts parallelism with the layer shape.

Pattern: early layers favor Y/X, the middle layer uses a Y/K/C three-level mapping, and late layers move toward C/K.

Early layer: Layer 1: Dim.: [64 3 224 224 7 7]

Dim.: [K C Y X R S]

Y	S	X	C	R	Y	K	C	S	R	K	Y	X	C
-	5	222	3	1	4	45	2	2	4	20	1	1	2

Med layer: Layer 8: Dim.: [128 64 56 56 1 1]

Y	C	R	K	S	X	Y	K	K	Y	C	S	R	X	C	R	K	X	S	Y	C
-	38	1	67	1	22	16	67	1	15	37	1	1	17	1	1	1	8	1	1	1

Late layer: Layer 18: Dim.: [512 256 14 14 1 1]

C	S	C	Y	K	R	X	K	X	S	Y	C	K	R
-	1	150	8	475	1	8	168	7	1	7	149	1	1

Found mappings for early, medium, and late ResNet-18 layers

End-to-End Wins Persist

15

Cross-model result: on s3, GAMMA stays ahead after moving from layerwise search to end-to-end DNN pipelines.

Latency edge: versus the best fixed mapping, GAMMA is 7.5x faster on MobileNet-V2 and ShuffleNet, 10.2x on MnasNet, and 20.2x on ResNet50.

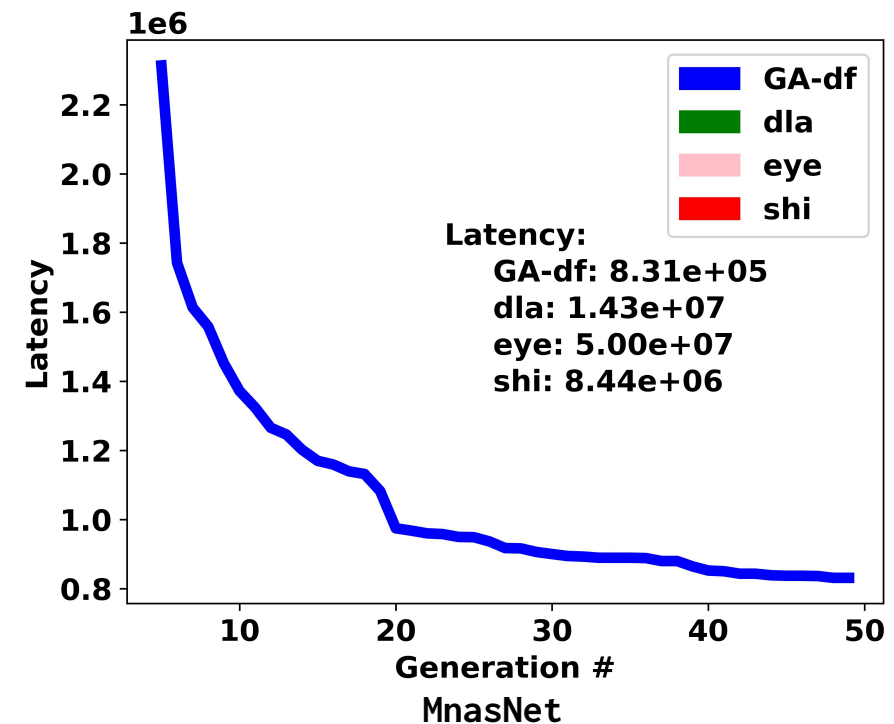
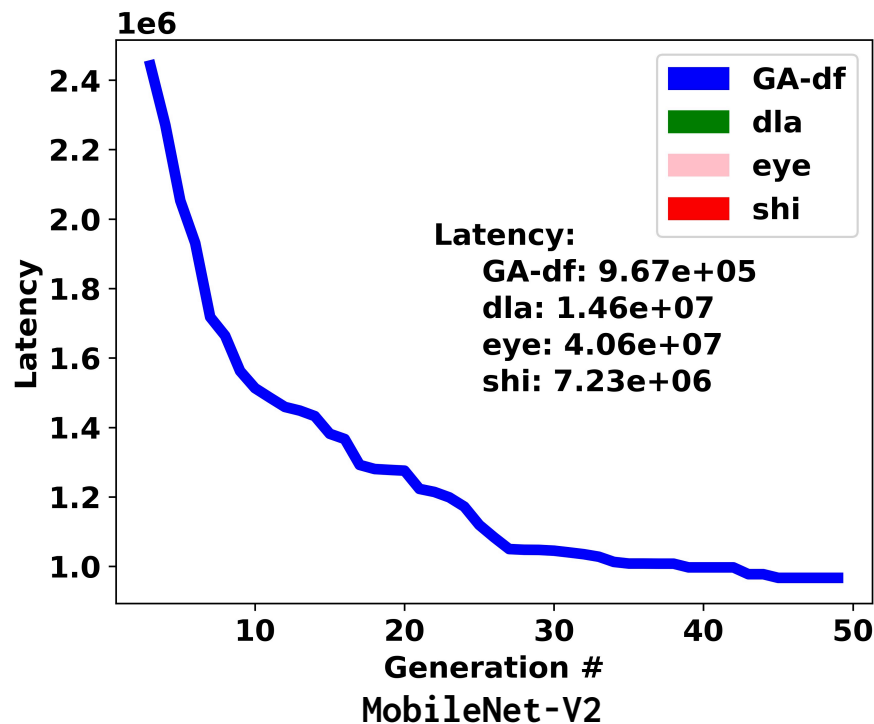
Model	Edge lat.	Cloud lat.	Edge en.	Cloud en.
MobileNet-V2	7.5x	5.0x	6.3x	2.0x
MnasNet	10.2x	29.0x	7.4x	2.1x
ShuffleNet	7.5x	18.4x	9.6x	2.2x
ResNet50	20.2x	75.8x	29.7x	1.9x

Convergence Is Fast

16

Sample efficiency: most of the latency drop arrives in the first 20 generations, well before the full 10K-sample budget is exhausted.

Examples: the search settles at $9.67\text{E}+05$ cycles on MobileNet-V2 and $8.31\text{E}+05$ on MnasNet under s3 Edge.



Pipeline Slack Can Be Reinvested

17

Extension: a second GAMMA pass preserves the bottleneck latency while relaxing non-critical layers for lower power or energy.

System	Latency-Power	Latency-Energy
ResNet-18	95% lower power at the same pipeline latency	58% lower energy at the same pipeline latency
VGG16	99% lower power	78% lower energy
Bottleneck note	Layer 2 sets the ResNet-18 pipeline latency	Stage 2 keeps other layers below that latency

Takeaway: once the bottleneck is fixed, the remaining throughput slack can be converted into system-level efficiency.

Outline

1. Motivation
2. Design
3. Evaluation
- 4. Takeaways**

What This Paper Establishes

19

Systems thesis: mapping is a first-order accelerator design variable, not a minor compile-time knob.

Method thesis: the crucial contribution is the flexible representation and operator set that make full-space search feasible.

Evidence: the biggest gains appear exactly where generic baselines fail, namely under tight constraints and richer mapping flexibility.

Limits: the evidence is cost-model-driven and centered on CNN-style accelerators, so broader hardware and workload validation remains open.